

Lean/Six Sigma Application to the Presidential International Travel Fellowship Process

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Abstract

The Education Abroad office at The University of Oklahoma (OU) takes a long time to notify students on whether they are awarded the President's International Travel Fellowship (PITF) scholarship. The PITF scholarship is awarded to assist high-needs OU students with their travel expenses while studying abroad. Using the DMAIC process, we identified some potential root causes of extensive waiting periods within the scholarship award process, developed a solution to reduce processing times, implemented that solution, and developed controls to maintain reduced processing time for awarding OU students the PITF scholarship. The two primary sources of waste were identified as the sorting of high-needs students from low-needs students and excessive office communication wait time.

The OU Education Abroad office used Microsoft Excel automation tools and implementation of deadlines to reduce the processing time of scholarship awards. The Excel tools were instrumental in reducing the processing time of sorting high-needs students from low-needs students. Before improvement, an employee was required to manually inspect student information and sort high-needs students from low-needs students. This process was time consuming for the employee. Excel automation reduced this sorting time to only one minute with only a handful of mouse clicks. Additionally, deadlines were implemented to expedite the long wait times between office communication, which previously took an excessive amount of time with high variability. The excessive office communication wait times were a substantial portion of award processing time. Establishing deadlines resulted in significantly expedited communication. Adherence to the new deadlines and continued use of the Excel automation tools will ensure long-lasting benefits from process improvement.

Implementation of improvements occurred in July 2023. The improvements reduced the mean award processing time from 19 days, with wide variability, to 7 days with much less variability. The unimproved process cost was \$2,257.75. The improved process cost is \$796.88. The improvement process cost totaled \$240 in training. Upon project completion, we achieved award cycle savings of \$1,460.87 per round of awards, resulting in an ROI of 608.7% after the first round.

1 Define

1.1 Problem Statement

The Education Abroad office at the University of Oklahoma takes longer than expected to notify students of whether they are awarded the PITF scholarship.

1.2 Identify CTQ/P

This section of our phase covers critical-to-quality/performance in the areas of feedback necessary from both students abroad and the employees operating the scholarship program. The top key indicators of student satisfaction included wait time, communication, scholarship decision, and additional opportunities for growth.

We highlighted each of these facets into their own respective survey questions through Google Forms. Sending the survey out to students would help us receive feedback on wait time, satisfaction of communication with the PITF, their assessment on their final scholarship decision, and additional concerns they may have raised in this time.

The results from students abroad indicated opportunities for growth, such as better, clearer communication of the stipend they receive, clarity of how the entire process works, and the reason for the rejection of their application. An announcement of their scholarship amount prior to their program payments was a large critique as well, as around a quarter of applicants waited for over 4 weeks, and none had received their announcement in less than a week as highlighted by the program.

1.3 Performance Metrics

The metric we are looking to improve is the overall processing time of student applications for study abroad scholarships. Processing time begins after the scholarship application deadline and ends when students receive their award notification.

1.4 Project Objectives

This project’s main goal is to reduce study abroad scholarship processing time from a range of 7–40 days to less than 7 days.

1.5 Identify Customers and Suppliers of the Process

Table 1 summarizes the suppliers, inputs, process steps, outputs, and customers (SIPOC) of the PITF award process.

Table 1: SIPOC Diagram of the PITF Scholarship Award Process

Suppliers	Inputs	Process	Outputs	Customers
Financial Aid Office	TerraDotta application	1. Create Master Flights Key	Award notification	Students
Scholarship Coordinator	Excel data	2. Request IDs from Financial Aid Office	Customer satisfaction	Program Coordinators
Front Office Staff		3. Create query for term applicants		Shared Business Services Center (SBSC)
Associate Dean of Students		4. Compile master spreadsheet		
		5. New query for program info		
		6. Create new report for program data		
Review Committee		7. Prepare/send report to Dean’s Office		

1.6 Benefits and Value of Project

Cost benefits. Table 2 reports the distribution of original wait times to respond to applicants. The wait time is estimated to decrease to 40 hours (about 1 and a half days) after the system has been refined, ideally aligning 100% in the <7 days category. Table 3 gives the estimated cost of the current and future process.

Organizational benefits.

- Decreased cycle time. Reduction in process time will reduce wait time, allowing for optimization of cycle time.

Table 2: Original Distribution of Time to Respond to Applicants

Time to respond	Original wait time
<7 days	0%
7–14 days	36.4%
21–28 days	38.6%
>28 days	25%

Table 3: Estimated Cost of the Current and Future Process (Define-Phase Estimates)

	Time (minutes)	Cost / minute	Total cost
Estimated current	9,907	\$0.25	\$2,476.75
Estimated future	2,400	\$0.25	\$600.00
Savings per round of awards			\$1,876.75
Cost of improvement (16 hours)			\$240.00
ROI after first round			781.98%

2 Measure

2.1 Current State Performance Data

Refer to Appendix A for the round-by-round process data collected on the “as is” process.

2.2 Value Stream Map of “As Is” Process

Figure 1 maps the current-state flow of the award process, from the first query through notification of the SBSC.

2.3 Current State Performance Summary

Table 4: Summary of Current-State Process Data (Appendix A)

Statistic	Minutes
Average	9,031
Standard deviation	857

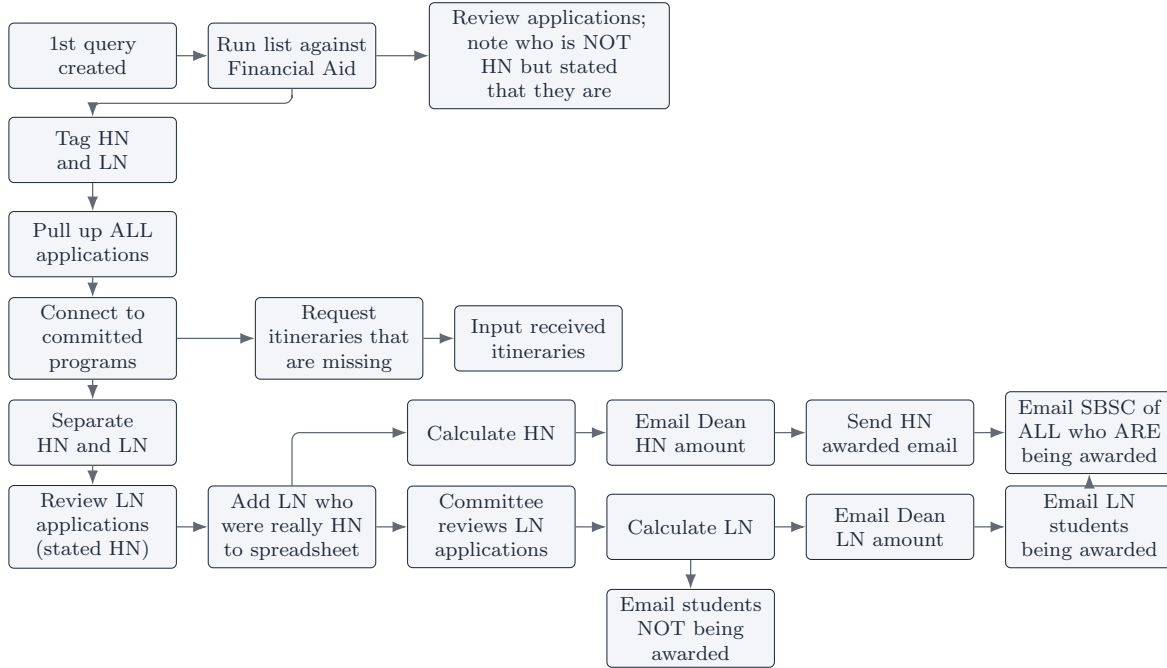


Figure 1: Value stream map of the “as is” PITF award process.

2.4 Develop a More Detailed Problem Statement

The Education Abroad office at the University of Oklahoma spends excessive time in notifying students on whether they are awarded the Presidential International Travel Fellowship because of the amount of time and effort it takes to verify students’ financial status, staff multitasking, tracking itineraries missing from programs, and waiting to hear back on approvals. This process is time-consuming and has a high amount of variability because of the dependence on receiving information from other sources.

3 Analyze

3.1 Potential Root Causes

The fishbone diagram in Figure 2 shows the many different obstacles in the process of communication with staff and students. This procedure requires multiple steps with often poor communication and inconsistent responses.

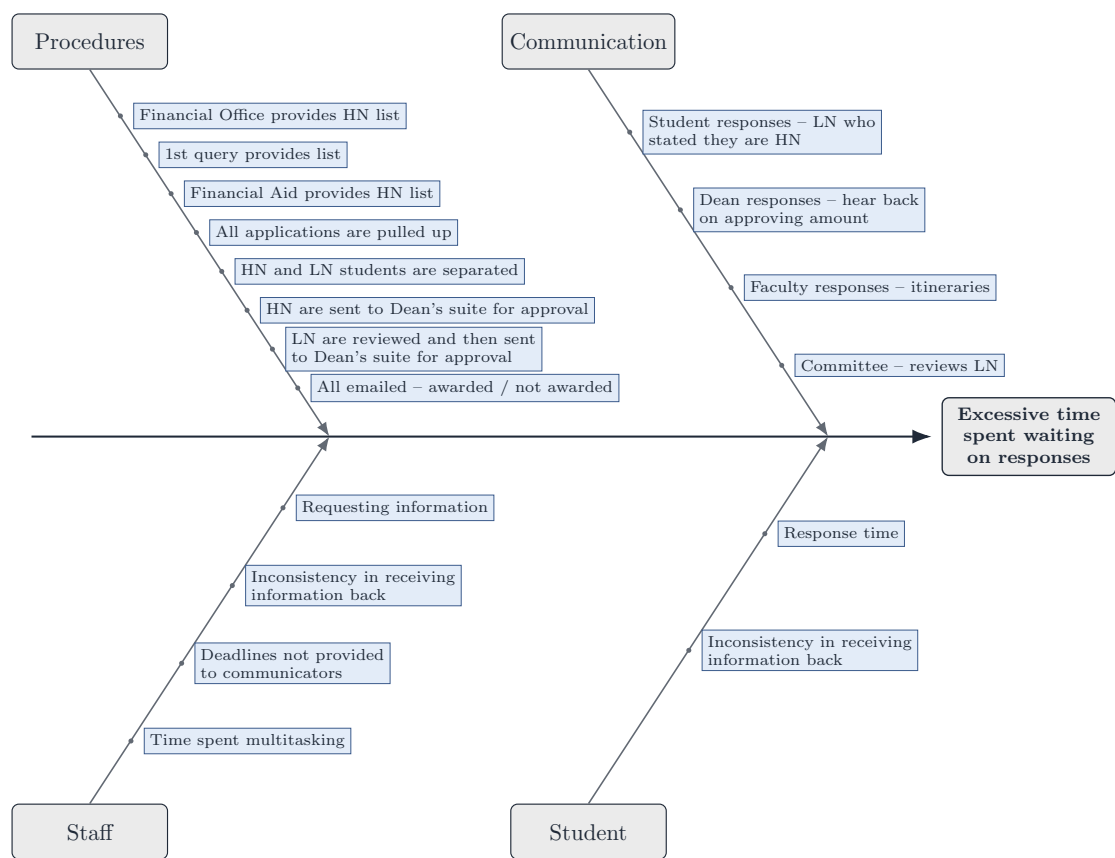


Figure 2: Fishbone (Ishikawa) diagram of potential root causes of excessive time spent waiting on responses.

3.2 Sources of Waste, Down Time and Other Issues

Throughout the PITF scholarship process there are many factors that play into the delays in students getting their results as to whether they have been awarded assistance to study abroad. As with any process with many moving parts and people, there are many communication delays and barriers that are making the process sub-optimal. One factor that is leading to the delayed process is the manual sorting of data to determine high-need and low-need students, which could be automated to increase productivity and increase response times to get the information into the hands of those who need it.

There is also the issue of faculty having to multitask on many issues and the PITF's lack of a deadline; this moves the PITF's work lower on the priority list of outstanding tasks. Another factor is that students believe they are high-need students when in reality they are low-need students. This discrepancy causes a delay in deciding who a high-need student is. The implementation of a deadline on both ends—for students and staff to provide accurate information and for faculty to be able to provide a response with that correct information—is another way down time can be reduced.

3.3 Verify and Prioritize Root Causes

We have chosen to address the process of time spent on receiving information in return and the time it takes to sort out applications in Excel as the root causes of wasted time in the PITF awarding process.

4 Improve

4.1 Decide on Tools for Fixing the Problem

The Education Abroad office is implementing Excel automation tools that will sort high-needs students and low-needs students. The office is also working within the department to finalize dates and deadlines for all parties involved to improve response times.

4.2 Map the Future State

Figure 3 presents the future-state value stream map with the automated sorting steps and consolidated communication with the Dean's office.

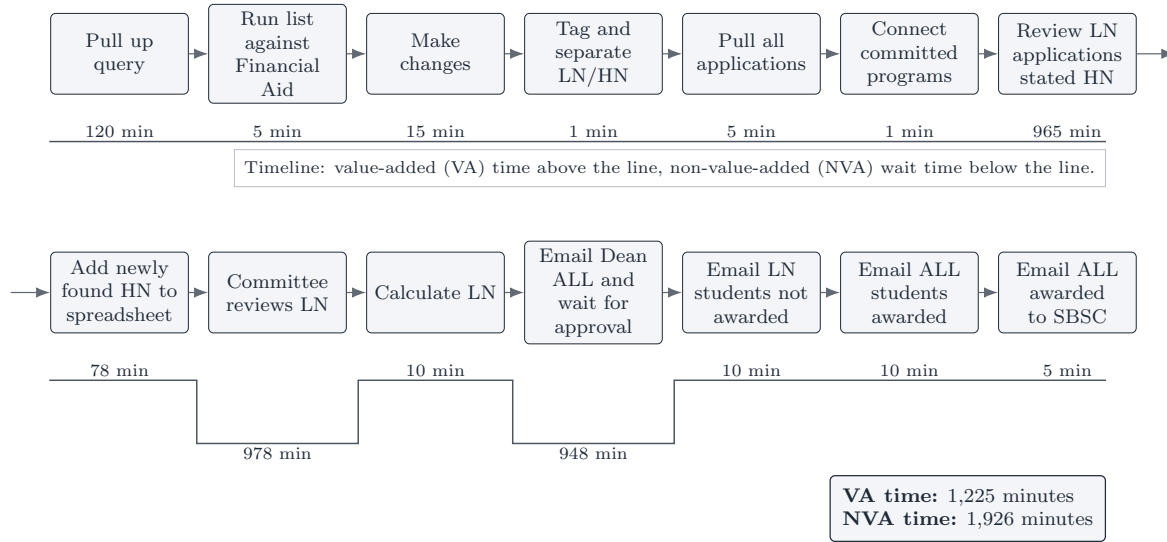


Figure 3: Future-state value stream map of the PITF award process. Value-added time: 1,225 minutes; non-value-added time: 1,926 minutes.

4.3 Improvement Plan

With the implementation of an automated spreadsheet, the Education Abroad office staff will input data related to the applications for Excel to sort into categories of high need and low need, eliminating the human factor that increases wait times. Additionally, deadlines will be implemented when requesting information or approval, mitigating wasted time. By utilizing deadlines and Excel sorting, the Education Abroad office will no longer waste time waiting on receiving information in return and time spent sorting.

4.4 Technology

Use of Microsoft Excel automation tools will be foundational to achieving process reduction in sorting high-needs students from low-needs students. No new technology is required to be purchased.

4.5 Pilot

A pilot was not needed.

4.6 Complete Items from Improvement Plan

The implementation of the Excel automation tools occurred in July 2023. With these tools, the person who was in charge of visually inspecting is now able to engage the automation

tools to achieve near-instant sorting; this eliminates time wasted by human sorting. Simultaneously, deadlines were put in place for those needing to provide information to the Education Abroad Office to continue the process. As a result, this will minimize time wasted waiting for responses. The rest of the process carries out as before.

4.7 Collect and Show Improvement

Refer to Appendix B for the round-by-round process data collected after the new tools were implemented. Table 5 summarizes the two data sets.

Table 5: Summary Process Data Before and After the New Tools Were Implemented

Statistic	Before new tools (min)	With new tools (min)
Average	9,031.0	3,187.5
Standard deviation	857.0	440.5

5 Control

5.1 Validate Performance Improvements

Table 5 is a side-by-side comparison of our summary values before and after the new tools were utilized as a part of our project.

Therefore, with the Excel tools and deadlines implemented, we were able to minimize the average process time by 64.7% and the standard deviation by 48.6%. This minimized process time from 19 to 7 workdays; more specifically, 5,843.5 minutes (excluding any wait times) have been reduced in total. ROI for the project was calculated to be 608.7% after the first round of awards.

5.2 Sustain and Control Plan

Continual usage of the Excel sorting and deadlines has eliminated much of the wasted time waiting for sorting and responses. Therefore, as our first step, the office will utilize the information they upload into the system, given the automation put in place. Next, set deadlines create urgency for any information needed, decrease the response time, and facilitate the process for high-need and low-need applicants.

5.3 Steps for Ongoing Improvements

An idea is to use this improvement in other areas of scholarships, not just PITF, to structure the method of using Microsoft Excel automation tools once the Education Abroad Office acclimates to its new method of selecting scholarship recipients. With the removal of the human factor integrated into determining high-need and low-need applicants, disbursement of money can be distributed based on the results to fulfill the needs of more high-need applicants in the future. This may additionally be used for a separate fund for respective emergencies abroad, although rare in some cases. This can include building off of the automation tools to allow for more flexible models of determining applicant strength.

5.4 Identify and Record Other Improvement Opportunities Discovered

Upon our completed work thus far, the means for continuous improvement for this project can take form in numerous different avenues for the betterment of the Education Abroad Office and scholarships as a whole. Firstly, that includes hiring an additional staff member to ease the workload of managing multiple applicants at once, and simplifying responsibilities for the scholarship team since a lot of current processes are automatic. Our next recommendation is to continue to reduce the time it takes to receive the scholarship once an applicant clicks ‘submit’ down to less than a week. One way to facilitate this would be to minimize the number of buttons used in the Excel functions for selecting ‘high-need’ and ‘low-need’ applicants. Currently there are 5 different buttons that can be reduced down to one for simplicity.

A Current State Performance Data

Table 6: Time per Process Step (Minutes) by Award Round, Before Improvement

Process step	1	2	3	4	5	6	7	8	9	10
Pull up 1st query	120	120	120	120	120	120	120	120	120	120
Run list against Financial Aid	5	5	5	5	5	5	5	5	5	5
Make changes to 1st query	10	15	20	15	15	15	20	15	15	15
Tag LN or HN	480	720	480	720	720	720	720	720	720	720
Pull up ALL applications	5	5	5	5	5	5	5	5	5	5
Connect to committed programs	720	960	480	720	480	960	480	960	480	600
Separate HN and LN	5	10	15	20	10	10	10	25	15	20
Request itineraries for any missing	1440	1920	1920	1440	3360	1890	2400	1780	2300	2010
Input received itineraries	100	240	100	180	80	120	120	120	120	120
Review LN applications stated HN	960	1320	1440	1880	1440	1440	1370	1320	1620	1800
Add LN who were really HN to spreadsheet	120	60	120	120	60	120	120	120	120	120
Calculate all HN	10	10	10	10	10	10	10	10	10	10
Email Dean with HN total and wait	1920	1250	1440	1440	1360	1500	1560	1750	1960	1200
Send email to HN students being awarded	240	240	240	240	240	150	180	180	240	200
Committee reviews LN applications	960	700	1440	1920	960	1000	1100	1040	970	990
Calculate all LN	10	10	10	10	10	10	10	10	10	10
Email Dean with LN total and wait to hear back	720	1120	1440	1440	110	900	1200	950	1130	1100
Email LN students NOT being awarded	10	10	10	10	10	10	10	10	10	10
Email LN students who ARE being awarded	10	10	10	10	10	10	10	10	10	10
Email list of ALL being awarded to SBSC	5	5	5	5	5	5	5	5	5	5

B Performance Data with New Tools Implemented

Table 7: Time per Process Step (Minutes) by Award Round, After Improvement

Process step	1	2	3	4	5	6	7	8	9	10
Pull up 1st query	120	120	120	120	120	120	120	120	120	120
Run list against Financial Aid	5	5	5	5	5	5	5	5	5	5
Make changes to 1st query	10	15	20	15	15	15	20	15	15	15
Tag and separate LN or HN	1	1	1	1	1	1	1	1	1	1
Pull up ALL applications	5	5	5	5	5	5	5	5	5	5
Connect to committed programs	1	1	1	1	1	1	1	1	1	1
Review LN applications stated HN	1440	720	800	970	1400	1350	1100	850	600	720
Add LN who were really HN to spreadsheet	60	120	60	60	120	60	120	120	60	60
Committee reviews LN applications	720	1000	1440	900	800	1200	1150	600	720	1250
Calculate all HN & LN	10	10	10	10	10	10	10	10	10	10
Email Dean with ALL total and wait to hear back	1440	1000	720	900	850	1150	600	720	1250	850
Email LN students NOT being awarded	10	10	10	10	10	10	10	10	10	10
Send email to ALL students that are being awarded	10	10	10	10	10	10	10	10	10	10
Email list of ALL being awarded to SBSC	5	5	5	5	5	5	5	5	5	5